Practice workbook · Apply digestive enzyme action to an unfamiliar situation

Novel contexts where digestion, or its absence, changes what is absorbed: oral hormones, PKU, chymosin, lactase loss.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-ENZYME-APPLIED-CONTEXT

1. Question 9 · part a.ii · 3 marks

January 2020 Paper 2

Worked-example retry · Attempt part a.ii only.

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9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

(3)

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1. Question 9 · part a.ii · 3 marks · Working and answer

Working and answer

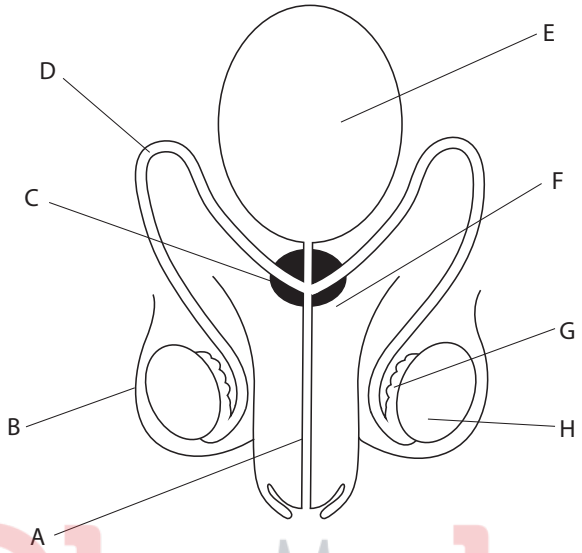
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2. Question 7 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

7 (a) The diagram shows the male reproductive system.



The table gives descriptions of structures in the male reproductive system.

Complete the table by giving a letter from the diagram to match each description.

Each letter may be used once, more than once or not at all.

(4)

Description of the structure	Letter
stores sperm	
produces sperm	
produces fluid to help sperm move	
contains cells dividing by meiosis	

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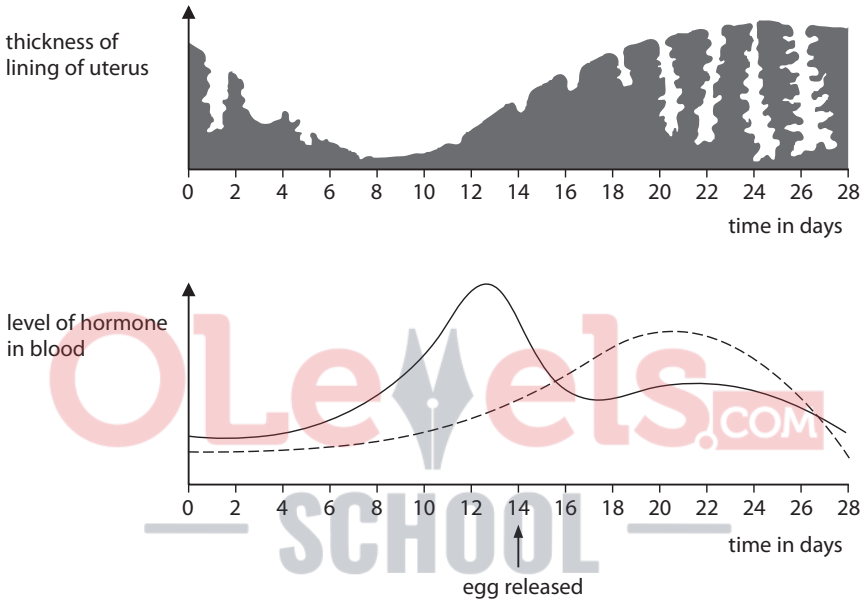
2. Question 7 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

(b) Changes occur in the female body during the menstrual cycle.

The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.



(i) Use information from the graphs to explain the cause of growth in the lining of the uterus after day 12.

(2)

(ii) Explain why fertilisation is less likely to occur from intercourse between days 8 and 12.

(2)

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2. Question 7 · part c.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part c.i only.

(c) A synthetic oestrogen called oestradiol is used to make contraceptive pills. Natural oestrogen, which is a lipid molecule, could be used but is less effective than oestradiol when taken as a pill.

(i) Suggest why natural oestrogen is not as effective as oestradiol when taken as a pill.
(2)

(ii) Give a reason why contraceptive pills only need to be taken for 21 days out of the 28-day cycle.
(1)

(Total for Question 7 = 11 marks)

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Working and answer

3. Question 7 · part a.i · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

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7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

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3. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 5 · part b · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

The number of people with lactose intolerance differs across the world. People in some countries are often tolerant because their ancestors have evolved tolerance because of their diet.

In a lactose intolerant person, any lactose taken into the gut will collect and ferment, forming methane and carbon dioxide plus other soluble products. These other soluble products lower the water potential in the gut.

Symptoms of lactose intolerance include diarrhoea, calcium deficiency and vitamin D deficiency.

(a) Suggest what is meant by the term disaccharide. (lines 3 to 4) **(2)**

(b) Give a reason why lactose is not absorbed by the epithelium. (lines 1 to 2) **(1)**

(c) (i) Explain what is meant by a gene being switched off. (lines 5 to 6) **(3)**

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4. Question 5 · part b · 1 mark · Working and answer

Working and answer

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5. Question 5 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Milk contains a carbohydrate called lactose. Lactose is not absorbed by the epithelium of the ileum or duodenum.

Lactose is a disaccharide which is broken down into glucose and galactose by the enzyme lactase. Babies produce large amounts of lactase but in some people the gene for producing lactase is switched off as they get older. When this gene is switched off people then become lactose-intolerant which means they cannot digest any lactose that is taken into the gut.

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5. Question 5 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

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(ii) Describe the effects on a person when the gene for producing lactase is switched off. (lines 3 to 7)

(2)

(d) Suggest how people in some parts of the world have evolved tolerance because of their diet. (lines 8 to 10)

(2)

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(e) Explain how the symptoms of lactose intolerance are caused. (lines 11 to 15)

(4)

(Total for Question 5 = 14 marks)

5. Question 5 · part c.ii · 2 marks · Working and answer

Working and answer

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6. Question 7 · part a.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.ii only.

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(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

6. Question 7 · part a.ii · 2 marks · Working and answer

Working and answer

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